

CogniCare

An AI-Based Cognitive Gaming & Memory Assistance Platform for Elderly Dementia Patients in the North Eastern Region

Idea Description · Smart India Hackathon

Problem Statement **SIH26003** · MDoNER · Software · MedTech / BioTech / HealthTech

1. Executive Summary

CogniCare is an AI-powered Cognitive Digital Therapeutics (CDTx) and memory-assistance platform purpose-built for elderly dementia, Alzheimer's, and Mild Cognitive Impairment (MCI) patients across the eight states of the North Eastern Region (NER) of India — a direct, working response to MDoNER Problem Statement SIH26003.

The platform combines a **comprehensive suite of culturally anchored serious cognitive games**, a **local on-device AI clinical-intake engine**, **multilingual voice-assisted interaction covering the major languages of the NER**, **real-time adaptive difficulty driven by a quantified multi-domain cognitive profile**, a **zero-touch QR kiosk**, **fully offline-first operation**, **caregiver monitoring with SOS surveillance**, and an **NER-wide command center for the Ministry**. It is engineered around the lived culture of the Northeast — tea gardens of Assam, Majuli pottery, Meghalaya's living root bridges, Brahmaputra boats, the Bihu dhol, and the region's folklore — because a therapy a grandmother recognises is a therapy she will use.

CogniCare is a **digital therapeutic**, not a brain game: every activity maps to a cognitive domain, every session feeds a quantified clinical profile, and each patient's trajectory is calibrated against standard neurocognitive scales. Caregivers see progress, clinicians see evidence, and MDoNER sees population-level epidemiology. It works where patients actually are — villages with poor connectivity and low specialist access — and no patient data ever leaves the local clinic or device.

2. Problem Deep-Dive and Regional Context

The demographic pressure. India has one of the world's largest and fastest-growing elderly populations living with dementia and age-related cognitive disorders — a burden projected to grow sharply as life expectancy rises.

The NER-specific crisis. The eight states — Assam, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, Arunachal Pradesh, and Sikkim — share three structural burdens:

1. **Geographic isolation** — hilly and riverine terrain, dispersed rural settlements, and long travel distances place families days away from any neurology or geriatric consultation.
2. **Severely limited specialists** — among India's lowest densities of neurologists, psychiatrists, and geriatricians, concentrated in a few urban pockets, leaving the rural majority without screening, therapy, or follow-up.
3. **Language and cultural diversity** — dozens of languages and distinct cultural identities mean any solution delivered only in English or Hindi excludes precisely the elders who need it most.

The human cost. Dementia in NER families manifests as memory decline, confusion, anxiety, depression, wandering, and social isolation for the patient — and as continuous, unresourced monitoring, fatigue, and burnout for caregivers, with a disproportionate burden on working-age women. Without structured cognitive stimulation and monitoring, decline accelerates, quality of life collapses, and families become de-facto care institutions with no support.

Why the status quo fails. Existing offerings fall into three unusable categories:

1. **English-only western brain games** with no cultural anchor, no regional language, and no clinical grounding — irrelevant to an Assamese or Mizo elder.
2. **Clinic-based clinical assessment** that requires a specialist visit — impossible for rural families to sustain.
3. **Static reminder apps and pill-timers** that fragment care, ignore the cognitive dimension, and give caregivers no visibility into improvement.

Between “too clinical to reach the patient” and “too trivial to help the clinician” there is a gap. CogniCare fills it.

3. The Solution: CogniCare CDTx

A complete, closed-loop Digital Therapeutics platform in five concentric layers:

1. **Clinical intake** — a caregiver uploads a patient’s clinical record; a hybrid pipeline (rule-based fast-path + local on-device LLM) extracts and quantifies a **multi-domain cognitive profile** — cognitive, functional (Instrumental ADL), and behavioural — and calibrates a severity baseline.
2. **Adaptive therapy** — a personalised therapy suite of serious cognitive games whose difficulty adapts every session.
3. **Daily living assistance** — voice-enabled reminders for medicines, hydration, routine activities, and appointments.
4. **Monitoring & analytics** — caregiver dashboards, progress tracking, trajectory views, and an alert system.
5. **Public-health intelligence** — a state-wise command center visualising adherence, stability, and adoption.

Every layer is **offline-first**, **privacy-preserving**, and **accessibility-engineered for elders**.

4. The Cognitive Gaming Suite — Meeting Requirement (a)

CogniCare’s therapy portfolio spans the full cognitive spectrum the problem statement demands, organised into clinically recognised domains:

- **Memory improvement & daily routine recall** — face- and event-based recall games, AI-guided conversational reminiscence, chronological life-milestone activities, family photo puzzles, auditory memory games, a structured daily-life story journey, and prospective-memory exercises built around the patient’s actual day. **Attention & concentration** — selective-attention tasks, auditory working-memory span, spatial navigation and tracking, visual discrimination and search, and spatial orientation activities. **Pattern and object recognition** — pattern completion, categorisation and sorting, free-form drawing, and tactile motor-construction tasks. **Emotional and mental engagement** — immersive 3D spatial scenes, rhythm-and-tempo entrainment,

motion-interactive movement games, and a conversational AI companion that grounds the patient emotionally in their own language.

Each game is mapped to a clinical domain so therapy is **prescribed stimulation** targeted at the patient's specific deficits — never random entertainment.

5. Adaptive AI Difficulty Engine — Meeting Requirement (b)

Every session feeds a live adaptive engine:

- **Multi-domain baseline** — the clinical-intake pipeline converts uploaded records into per-domain severity levels, so a patient never starts at the wrong difficulty. **Scaffolding** — hints and progressive support that withdraw as performance improves, following established rehabilitation principles. **Spaced retrieval** — memory items reappear at personally optimised intervals to maximise consolidation. **Continuous biomarkers** — movement, response timing, and voice patterns surface motor and attention signals that automatically tune activities. **Fatigue and hesitation detection** — subtle timing and interaction cues prompt rest or a domain switch. **Trajectory prediction** — longitudinal performance projects cognitive trends and raises early-intervention cues for clinicians. **Personalised pacing** — speech and interaction tempo calibrated to the patient's severity.

CogniCare measures each patient against **their own previous self** — yesterday's performance vs. today's — which is clinically sound and emotionally constructive.

6. Multilingual and Voice-Assisted Interaction — Meeting Requirement (c)

CogniCare is **multilingual-first**, covering the major languages spoken across the NER — Assamese, Bengali, Hindi, Nepali, Meitei (Manipuri), Mizo, Khasi, Bodo, Garo, and more — designed so additional languages can be added without architectural change.

- **No blank screens** — a fallback-safe translation architecture ensures that even untranslated content degrades gracefully to a primary language instead of rendering empty. **Full voice assistance** — every instruction, memory, reminder, and the AI companion can be spoken aloud through a hybrid neural-audio + native speech engine with cascading voice fallbacks, guaranteeing a comprehensible voice even where no native TTS model exists. **Read-aloud by default** — texts can be spoken automatically for elders with low literacy or visual impairment. **Conversational interaction** — voice-driven navigation and a spoken companion for patients who cannot read or type.

7. Cultural Anchoring in NER — Meeting Requirement (d)

CogniCare is built **from** Northeast culture rather than translated **at** it:

- **Themes** — Assam tea gardens, Majuli pottery, Meghalaya's living root bridges, Brahmaputra river crossings, the hornbill, the Bihu dhol. **Craft & motifs** — traditional weaving patterns, floor-art drawing, heritage recipes, and culturally familiar objects for memory stimulation. **Sound** — native instrumental and percussive audio, including rhythm entrainment for motor rehabilitation. **Folklore & language** — AI-generated local proverbs, tales, and life-story chapters for reminis-

cence. **Personal life-material** — caregivers upload the patient's own family photographs and places, so therapy is built around **their** life in **their** language.

For a dementia patient, familiarity is therapeutic. This is the engagement insight that makes the platform work where generic games fail.

8. Daily Living Assistance & Reminders — Meeting Requirement (e)

- **Medicine reminders** — timed, spoken, acknowledge-confirmed alerts per medication schedule. **Hydration reminders** — gentle periodic prompts. **Daily activities** — routine scheduling with prospective-memory activities that rehearse the patient's real day. **Medical appointments** — reminders synced to caregiver-created events. **Extended reach** — reminder delivery can extend beyond smartphones (e.g., interactive-voice-response relay) for households with basic phones. **Adherence tracking** — acknowledgments flow back to the caregiver as data.

9. Caregiver Monitoring & Dashboards — Meeting Requirement (f)

- **Cognitive performance tracking** — per-domain progress, session history, difficulty trajectory. **Adherence analytics** — activity completion vs. prescription, reminder acknowledgment rates. **Biomarker & trajectory views** — intuitive charts of cognitive and physical signals over time. **Alert system** — missed-medication, geofence, vitals, and SOS escalations; declining-trend cues that suggest clinician review. **Secure patient identity** — a printable QR identity card aligned with national digital-health conventions (ABDM/ABHA), and a one-touch SOS raising a caregiver alert in real time.

10. Offline-First, Low-Connectivity Architecture — Meeting Requirement (g)

- **On-device intelligence** — the clinical AI and adaptivity run locally; core therapy has no cloud dependency. **Native app** — the platform is packaged as a native Android tablet/mobile app with hardware bridges (camera, text-to-speech, haptics, screen keep-awake kiosk mode, safe back-button handling) and pre-seeded local assets. **Local-first storage & LAN sync** — a clinic-local server keeps all records on-site, syncing only within the clinic network. **Graceful degradation** — if speech or network services are unavailable the platform falls back seamlessly; even the login path supports a fully functional offline session.

This design means CogniCare is **more valuable the further it is from connectivity** — the opposite of typical cloud-AI health products.

11. Elderly-Friendly Mobile/Tablet Interface — Meeting Requirement (h)

- **Accessibility-first, GIGW-aligned** — multiple font sizes, high-contrast mode, and large tactile targets. **Kinesthetic design** — vibration feedback on taps compensates for tremor; oversized controls and generous spacing. **Zero-touch entry** — a QR kiosk removes passwords; a three-step flow leads to a spoken greeting. **Form-factor adaptive** — orientation and layout tuned for tablet and phone. **Deliberately calm** — full-screen distraction-free therapy, safe exit handling, legibility-first typography, and severity-calibrated speech pacing.

This is not a standard enterprise dashboard — it is a calm, dignified, one-thumb screen a 75-year-old can use on the first try.

12. Patient-Data Security & Privacy

- **Privacy-first architecture** — biometrics, voice, and camera data are processed on-device; clinical AI never transmits patient health information. **National digital-health alignment** — QR identity and kiosk flows are designed for Ayushman Bharat Digital Mission (ABDM/ABHA) compatibility. **Secure sessions** — token-based authentication with expiry, idle auto-timeout, and role-separated patient/caregiver/administrator surfaces. **Scoped transport** — any local-network traffic is limited to the clinic’s own server; no external patient-data route exists.

13. Technology Approach

Layer	Approach
Web frontend	Modern React-based stack, typed, multilingual-first, accessibility-hardened
3D / vision / QR	GPU-accelerated 3D, computer-vision hand tracking, camera QR scanning
AI / LLM	Local on-device models for clinical extraction and conversational reminiscence
Backend	A robust Java/Spring-based REST service with secure auth and media handling
Data	Clinic-local relational storage, demo-friendly alternative, multi-patient schemas with family and biomarker history
Native app	Capacitor-packaged Android build exposing hardware capabilities

The entire platform builds and verifies reproducibly from source across all layers.

14. Innovation: What Makes This Different

1. **Culture-native dementia therapeutic for the NER** — not a translated western app, but built from NER cultural material.
2. **Closed clinical loop** — intake → adaptive therapy → monitoring → epidemiology in one offline platform.
3. **Affordable continuous cognitive assessment** — replacing episodic, unreachable specialist screening with on-device quantified profiling.
4. **Geriatric-grade accessibility** — accessibility thinking applied deliberately to elderly motor, vision, and cognitive constraints.
5. **Ministry-scale population intelligence** — the same device that treats a grandmother also streams de-identified data to a state-wise command center, giving MDoNER its first NER-wide dementia-telemetry gauge.
6. **True offline intelligence** — on-device AI and clinic-LAN sync engineered for connectivity-scarce geographies.

15. Feasibility & Execution Plan

The platform already exists as a **working prototype** across all three layers (frontend, backend, native). Planned phases:

Phase	Focus
P1 — Core therapeutics	Game portfolio, adaptive engine, multilingual + voice layer
P2 — Clinical pipeline	Record intake, quantified multi-domain profiling, severity calibration, local AI
P3 — Caregiver & monitoring	Dashboards, reminders, alerts, SOS, surveillance
P4 — Offline & native	Android packaging, offline sync, kiosk mode, clinic-local deployment
P5 — Ministry telemedicine	Command center, administration, audit, digital-health alignment

Each phase ends in verifiable builds, end-to-end offline smoke tests, and prototype demos.

16. Scalability & Adoption Roadmap for MDoNER

— **Pilot** — deploy to selected PHCs across two NER states with a cohort of elderly patients; instrument adoption, adherence, and outcome proxies. **Widen** — onboard Anganwadi/ASHA workers, community kiosks in village councils and PHC waiting halls, and reminder relay for non-smartphone households. **Scale** — roll out across the eight states; the command center becomes a standing health-intelligence surface informing NER dementia policy and screening drives. **Extend** — caregiver co-pilot support, tele-health integration, and physical kiosks for resource-constrained centers.

17. Expected Impact

— **Clinical** — early cognitive intervention for elders currently unreachable; measurable maintenance/improvement of cognition; reduced anxiety and isolation through engagement and companionship. **Caregiver** — structured continuous monitoring that relieves unmanaged strain; even caregiver-burnout signals are surfaced to administrators. **Systemic** — the first NER-wide dementia adherence and epidemiology dataset, giving MDoNER and allied health schemes an evidence base for allocation. **Equity** — multilingual, offline, low-cost delivery to the most isolated elderly — inclusive digital healthcare by definition.

18. Conclusion

CogniCare addresses every element of SIH26003 with a **working, offline-first, multilingual, AI-adaptive digital therapeutic** engineered for the Northeast. It turns an isolated, anxious, declining experience into a dignified, engaged, clinically measured one — for the grandmother in a Shillong village, the exhausted caregiver in Golaghat, the overstretched clinician in Guwahati, and the Ministry that must see the whole region. Not a generic app for the Northeast — a therapeutic built **by and for** the Northeast.